

Restoring M1/M8 Numerical Agreement in RDNA4 Speculative Decoding

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Abstract

A pinned Radiance/vLLM inference stack produced different target logits when the same saved token history was processed one position at a time (M1) or eight positions at a time under D7 speculative verification (M8). The investigation localized numerical differences in GDN prefill and decode, convolution, gated normalization, attention, residual normalization and the BF16 vocabulary head. The repaired, compiled implementations now agree on top-1/10/20 membership, ordering, retained scores, inclusive boundary ties and full-vocabulary logit hashes at 10,000 decode positions and 23 initial-prefill predictions. Four subsequent performance changes reduced a repaired verification round from approximately 76 ms to 60 ms in the measured 60K-input control.

This report documents an empirical result on a pinned configuration. It does not establish arbitrary-input equivalence, equality to an independently specified mathematical model, or improved task accuracy. The separate eager-versus-compiled reference discrepancy is reported below.

1. Configuration and reference

- One AMD Radeon AI PRO R9700, gfx1201, TP1.
- Radiance 1.0.16 / vLLM 0.28, PyTorch 2.12.0, ROCm 7.14, Triton 3.7.1.
- Qwen3.8-27B-Uncensored-MXFP4-awq; 64 layers, including 48 GDN and 16 attention layers; hidden width 5,120; vocabulary size 248,320.
- The target vocabulary head uses the full original BF16 weights in both arms. INT2 candidate selection is excluded from the equality claim. Other model weights, activation quantization and cache formats remain part of the pinned quantized backend; this is not an unquantized BF16-model comparison.
- The compiled comparisons use Inductor and piecewise GPU graphs, with capture sizes [1, 2, 4, 8]. Actual target graph replay is audited.
- The numerical reference is the explicitly selected serial implementation. Repairs also affect M1/preparation paths, so repaired M1 is measured afresh.

The 10K corpus comprises 23 saved Pi continuations with 57,008–65,527-token prefixes. Both arms consume identical saved token histories at aligned logical positions. Each M8 group processes eight positions with all seven supplied proposals accepted. The denominator is 10,000 processed decode positions, excluding intermediate-prefill head calls. Final initial-prefill predictions have a separate denominator of 23. No generated tools execute during replay.

2. Agreement definitions

For each position t , let $z1(t)$ and $z8(t)$ be the full vocabulary logit vectors. T_k sorts logits by decreasing score and then increasing token ID for ties.

```
set agreement(k)  = count(set(T_k(z1)) == set(T_k(z8))) / N
order agreement(k) = count(T_k(z1) == T_k(z8)) / N
mean shared(k)    = sum(|set(T_k(z1)) intersect set(T_k(z8))|) / N
```

Retained-score equality, inclusive boundary-tie membership and full-vector SHA-256 equality are separate checks. Matching hashes are strong empirical evidence, not a collision-free mathematical proof. Agreement measures execution consistency; it is not a percentage of questions answered correctly.

3. Top-1/10/20 and ordering results

Final compiled, optimized repair: full 10K corpus

Prediction	Same token set	Same ordering	Mean shared tokens
Top 1	10,000 / 10,000 (100%)	10,000 / 10,000 (100%)	1 / 1
Top 10	10,000 / 10,000 (100%)	10,000 / 10,000 (100%)	10 / 10
Top 20	10,000 / 10,000 (100%)	10,000 / 10,000 (100%)	20 / 20

Retained scores, inclusive tie sets and full-vocabulary hashes also match at all 10,000 positions. All measures separately match for **23/23 initial-prefill predictions**. Both workers exited successfully. The completion audit checked all per-continuation receipts, fixture identities, widths, aligned positions, runtime receipts and the recomputed aggregate.

Compiled before/after control: 320 positions

This is the directly comparable compiled control, separate from the larger historical eager run. Each implementation is compared with its corresponding M1 reference.

Prediction	Original M8: same set	Original M8: same order	Fixed optimized M8: same set	Fixed optimized M8: same order
Top 1	318/320 (99.375%)	318/320 (99.375%)	320/320 (100%)	320/320 (100%)
Top 10	161/320 (50.3125%)	17/320 (5.3125%)	320/320 (100%)	320/320 (100%)
Top 20	76/320 (23.75%)	0/320 (0%)	320/320 (100%)	320/320 (100%)

Full-vocabulary hashes match at 0/320 original positions and 320/320 repaired positions. The original initial-prefill vectors also differ. Original compiled M1 versus repaired compiled M1 agrees on top-1 at 319/320 positions and differs in every full-vector hash; reference changes are material to this experiment.

Historical unfixed eager baseline: same 10K corpus

Prediction	Same token set	Same ordering	Mean shared tokens
Top 1	9,852/10,000 (98.52%)	9,852/10,000 (98.52%)	0.9852 / 1
Top 10	4,857/10,000 (48.57%)	799/10,000 (7.99%)	9.3487 / 10
Top 20	2,270/10,000 (22.70%)	9/10,000 (0.09%)	18.6781 / 20

These historical results use eager execution. They must not be presented as an unfixed **compiled** 10K measurement. The final compiled run uses a fresh compiled M1 reference, not this saved eager reference.

4. Every compiled GPU stage/group, with repairs

Both profiles use the same 60K-input Pi fixture, compiled execution and actual piecewise graph replay. The table sums GPU kernel durations over eight profiled rounds and divides by eight. It covers every recorded kernel dispatch exactly once. Target layer stages are totals across all layers per round, not per layer. The full BF16 head is used on both sides.

Some original operations were fused by Inductor, so separate times for their constituents do not exist in the captured profile. Those operations retain a shared timing group. Matrix projections are likewise grouped by actual kernel dispatch rather than assigning unsupported timings to individual projections. `kernel1-dispatches.csv` contains every kernel name, group, call count and time.

Compiled GPU stage/group	Old M8 (ms)	Fixed M8 (ms)	Repair / interpretation
MXFP4 folded projections	24.243	25.496	No projection arithmetic change.
MXFP4 projections, split-4 kernel	7.633	7.554	No projection arithmetic change.
MXFP4 projections, split-1 kernel	4.981	4.973	No projection arithmetic change.

Compiled GPU stage/group	Old M8 (ms)	Fixed M8 (ms)	Repair / interpretation
Attention decode	4.104	5.673	Keep each query's serial causal tile/reduction contract; share KV reads across eight queries.
Attention split-KV merge	0.255	0.124	Preserve the serial per-query split/merge arithmetic.
Activation FP8 quantization	2.485	0.815	Same quantization kernel; measured timing change is not evidence of a quantizer repair.
GDN recurrent update and gates	1.150	1.211	Match serial gate precision, reduction order, recurrent transition and output rounding; preserve packed QKV views.
GDN convolution	0.769	0.228	Use the qualified serial product/accumulation order with rolling state; remove unnecessary packed-QKV copies.
Attention KV write	0.148	0.047	No KV-format change; timing includes the same cache-write kernel.
Explicit residual and Q/K/final normalization	Included in fused group	0.813	Preserve reduction tree, rsqrt and BF16 rounding boundaries; retain FP32 residual sums in registers.
Explicit GDN gated normalization	Included in fused group	0.098	Fix the row tile at the serial shape while processing all rows in parallel.
Compiler-fused normalization, activations, embedding and other pointwise work	1.398	0.336	Bind repaired normalization as opaque compiled operators before graph capture; remaining operations retain compiler fusion.
Target copies, conversions and buffer initialization	0.633	0.359	Keep packed GDN views and remove three materializations plus concatenation; other copies remain.
Full BF16 vocabulary head	4.106	4.134	Two interleaved M4 groups in one HIP launch preserve serial arithmetic without two launches and concatenation.
Drafter, including proposal head	6.334	6.447	Unchanged drafter; kept separate from target correctness.
GPU work outside attributed model stages	0.918	0.512	Includes sampling/state bookkeeping/copies; no invented per-operation attribution.

Target-body subtotal: **47.798 ms old; 47.726 ms fixed**. This subtotal includes the target rows above and must not be added again. Total recorded GPU kernel duration: **59.156 ms old; 58.819 ms fixed**. Kernel durations can overlap and exclude host/queue gaps; they are not the wall-clock round latency.

Changes in timings of unchanged kernels, especially activation quantization, are observations from these short profiles. They do not establish a causal speedup from a code change in that kernel. The repair does not make every stage faster: attention remains more expensive than the original arithmetic, while other savings recover the total round cost.

Prefill has a separate causal/chunk-partition repair and is outside this steady decode profile. Its speed is not inferred from the decode table. Sampling, accept/reject bookkeeping and state metadata outside the attributed model scopes are retained in the final accounting group; CPU-only scheduling has no GPU-kernel duration in this table.

5. Unprofiled serving-speed controls

Compiled configuration	Median round	Measured output rate	Evidence scope
Original, full BF16 target head	59.619 ms	85.212 tok/s	Three natural responses, 2,518 output tokens
Correctness repair, before performance recovery	76.038 ms	68.612 tok/s	Three natural responses, 2,237 output tokens
Repair after first three performance changes	60.985 ms	83.907 tok/s	Same three repaired responses, byte-identical outputs
Final repair, all four performance changes	59.997 ms	86.665 tok/s	One 743-token natural response

The matching pre-optimization repeat for that final response was 76.025 ms and 69.687 tok/s. The output digest is unchanged. The final round is about 0.64% above the original full-BF16 control's three-repeat median, but the different repeat counts limit precision. The output rates also reflect acceptance, so they must not be interpreted as pure kernel speed ratios.

These direct-engine controls exclude HTTP/Pi, snapshot publication, cold prefill and warm-up. They are separate from the earlier 60K-generated-token verify-head study, which also used compilation. The old legacy INT2 block-8/rerank-80 control measured 56.982 ms and 89.128 tok/s here, but uses an approximate target head and is not the numerical reference for this report.

6. Repairs and upstream ownership

1. **GDN prefill:** raw per-token gates and an ordered transition avoid the diagnosed dependence on future gates, cumulative-gate rounding and chunk partition. Core implementation belongs with libr4d; adapters belong in Radiance.
2. **GDN decode:** match serial convolution arithmetic, gate precision, recurrent reductions, output rounding and retained-prefix state. Rejecting a speculative suffix must not make its state persistent. Packed QKV views remove copies.
3. **Gated normalization:** retain the serial row tile without serializing all eight rows. The vendored FLA dispatcher belongs in vLLM.
4. **Attention:** retain each query's causal range and serial reduction/split policy; share KV reads to recover performance. Core ownership is libr4d.
5. **Residual/QK/final normalization:** preserve the chosen finite-precision boundaries; retain FP32 residual sums in registers. Existing vLLM rounding PRs must be checked for overlap before proposing another repair.
6. **Full BF16 head:** interleave two arithmetic-preserving M4 groups in one HIP launch. This avoids the first repair's duplicated launch/concatenation overhead. The upstream skinny-GEMM implementation belongs in vLLM.
7. **Compiled integration:** install dispatch before tracing/capture, use opaque bindings where necessary, and validate startup-only fallbacks separately from real requests. Integration and conformance adapters belong in Radiance.

These are submission scopes, not a claim that the pinned adapters apply unchanged to current upstream main. The new ports require their own tests. Upstream links and submission state are maintained in `submissions.md`.

7. Remaining discrepancies and limits

Corrected eager M1 versus corrected compiled M1 has **98.11% top-1 agreement** on the same 10K corpus. Top-10 set/order agreement is 48.45%/7.56%; top-20 set/order agreement is 22.27%/0.04%. Every full-vector hash differs, including all 23 prefills. This separate discrepancy is unresolved and is not evidence that either execution is an independently proved mathematical reference.

The final 10K replay covers the all-seven-accepted D7 path. Separate small operator tests exercise acceptance boundaries, state/history comparisons, graph replay and injected faults, but this report does not claim full-model coverage of every rejection width, arbitrary input, long-context range, concurrency, snapshot restore or hardware platform. It does not prove that model-generated loops are eliminated or measure improved coding accuracy.

Native correctness is tied to source, binary and configuration identities. Unsupported changes require requalification. A replay corpus and matching hashes cannot establish a universal theorem about all future optimizations.

8. Evidence and reproduction

The evidence/ directory contains aggregate comparisons and kernel profiles, with file SHA-256 values in `evidence-sha256.json`. Recompute the stage table, kernel accounting and aggregate assertions without GPU use:

```
uv run python reports/d7-rdna4-2026-09-17/build_report.py
```

The original Pi transcripts, token IDs, ranked token IDs and raw traces remain private. The aggregate measurements can be audited from the published receipts; independent end-to-end reproduction needs a public replacement workload. Synthetic operator regressions accompany the corresponding source submissions.

Pinned arithmetic repair SHA-256: 8deef78cd58b355d85d142caac58a88ea68ad576a31fe7769cc3db49b18235e9. Final performance bundle SHA-256: d5e74aedd327d5543a0f4a47aeff80ec853c3eea5f9de49051b2d58ec109068. 10K summary seal: 740f52b749130d91ecb870153a02e8b28c232489c87997423146c2af2d5496f8.

References and attribution

- [Radiance](#), the qualified downstream stack.

- [libr4d](#), RDNA4 attention and recurrent kernels.
- [vLLM](#), including ROCm skinny GEMMs and vendored FLA.
- [Flash Linear Attention](#), serial GDN/normalization foundations.
- [vLLM batch invariance](#), related existing work.
- [PyTorch numerical accuracy](#), batched/reduction-order limitations.
- Existing normalization work: [vLLM #49639](#) and [#52243](#).

This contribution builds on those implementations. Batch invariance itself is not introduced by this report. Earlier DFlash RNG work in Radiance PR #8 backports [vLLM #54282](#) and is not presented as a new upstream discovery here.